Recap: Monotone Convergence Theorem $\rightarrow$ (a_n) is bounded and monotone, it must converge.

Importantly, the MCT doesn't always allow us to identify the value a given sequence converges to.

Ex: Known, $\sum_{n=1}^{\infty} \frac{1}{n^2} = 1 + \frac{1}{4} + \frac{1}{9} + \dots = \frac{\pi^2}{6}$.

$\sum_{n=1}^{\infty} \frac{1}{n^3} = 1 + \frac{1}{8} + \frac{1}{27} + \dots = ???$

Theorem 7.5: Sequences and Suprema

Let $S \subset \mathbb{R}$ (Subset) be a non-empty subset bounded above and let $\alpha \in \mathbb{R}$ be an upper bound for S. Then $\alpha = \sup(S)$ iff there exists a sequence (a_n) with $a_n \in S$ and $\lim(a_n) = \alpha$.

Proof: ($\Rightarrow$) Let $\alpha = \sup(S)$. Then for every $n \in \mathbb{N}$, it follows that there exists an $a_n \in S$ such that

$$\alpha - \frac{1}{n} \leq a_n \leq \alpha.$$

This defines a sequence (a_n) and since $\lim(1/n) = 0$ it follows by the squeeze theorem that $\lim(a_n) = \alpha$.

($\Leftarrow$) Suppose there exists a sequence (a_n) such that $a_n \in S$ and $\lim(a_n) = \alpha$. Then, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ with

$$|a_N - \alpha| < \varepsilon$$

$$\Rightarrow -\varepsilon < a_N - \alpha$$

$$\Rightarrow \alpha - \varepsilon < a_N.$$

Hence by analytic properties of the supremum, it follows that $\sup(S) = \alpha$. ☐

Ex: Let $A = \{1 - 1/n : n \in \mathbb{N}\}$. Clearly $A \neq \emptyset$ and $1 > 1 - 1/n$ for all $n \in \mathbb{N}$. So, 1 is an upper bound. Let (a_n) be the sequence $a_n = 1 - 1/n$. Clearly $a_n \in A$ for all $n \in \mathbb{N}$ and $\lim(a_n) = 1$, hence $\sup(A) = 1$. ☐

8 - Subsequences and the Bolzano-Weierstrass Theorem

Consider the sequence

$$(1, 2, 3, 4, \dots).$$

A subsequence could be $(1, 3, 5, 7, 9, \dots)$ or $(2, 4, 6, 8, \dots)$.

Defn: (Subsequence)
Let (n_k) be an increasing sequence of positive integers.

That is,

$$n_1 < n_2 < n_3 < \dots$$

for each $n_k \in \mathbb{N}$. Then, given a sequence (a_n), the sequence (a_{n_k}) is called a subsequence of (a_n).

Ex: (i) Let (a_n) be the sequence $a_n = (-1)^n$.

Defining $n_k = 2k$ with $k \in \mathbb{N}$. Clearly $(n_k) = (2, 4, 6, \dots)$.

Then, the subsequence (a_{n_k}) is given by

$$(1, 1, 1, \dots).$$

(ii) Let (b_n) be the sequence $b_n = 1/n$. Let (n_k) be the sequence

$$(n_k) = (10, 100, 1000, \dots).$$

Then $$(a_{n_k}) = \left(\frac{1}{10}, \frac{1}{100}, \frac{1}{1000}, \dots\right)$$

Theorem 8.3: Convergence and Subsequences

A sequence (a_n) converges to $a \in \mathbb{R}$ iff every subsequence of (a_n) converges to a.

Proof: ($\Rightarrow$) Suppose (a_n) converges to $a \in \mathbb{R}$ and let (a_{n_k}) be a subsequence of (a_n). Then, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t

$$|a_n - a| < \varepsilon$$

for every $n > N$. Notice that $n_k \geq k$ for each $k \in \mathbb{N}$. Hence, $k > N$ we have that $(n_k \geq k > N)$

$$|a_{n_k} - a| < \varepsilon. \quad \square$$

($\Leftarrow$) Exercise

Corollary 8.4 : Divergence and Subsequences

If there exists two distinct subsequences of a sequence (a_n) that converge to different values, then (a_n) must be divergent.

Ex: Let (a_n) be the sequence given by $a_n = (-1)^n$. Let (n_k) be the sequence $n_k = 2k$ and $(\tilde{n}_k)$ be the sequence $\tilde{n}_k = 2k-1$, for $k \in \mathbb{N}$. Then

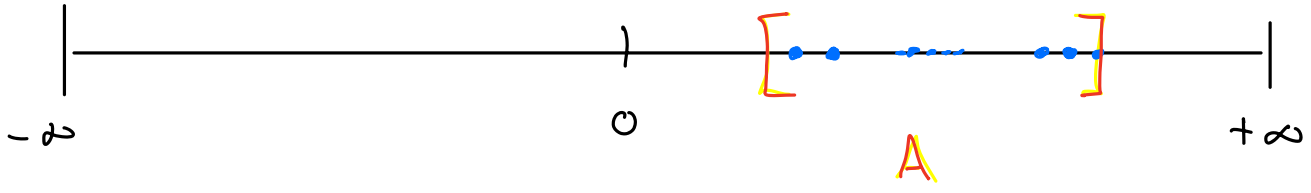
$$(a_{n_k}) = (1, 1, 1, 1, \dots)$$

and

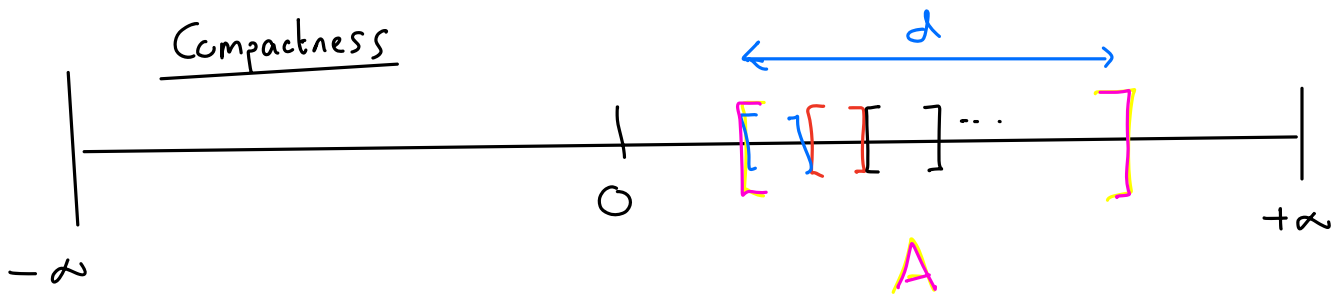
$$(a_{\tilde{n}_k}) = (-1, -1, -1, -1, \dots).$$

Clearly $\lim(a_{n_k}) = 1 \neq -1 = \lim(a_{\tilde{n}_k})$. Thus (a_n) is divergent. ☐

Pic :



Pic:

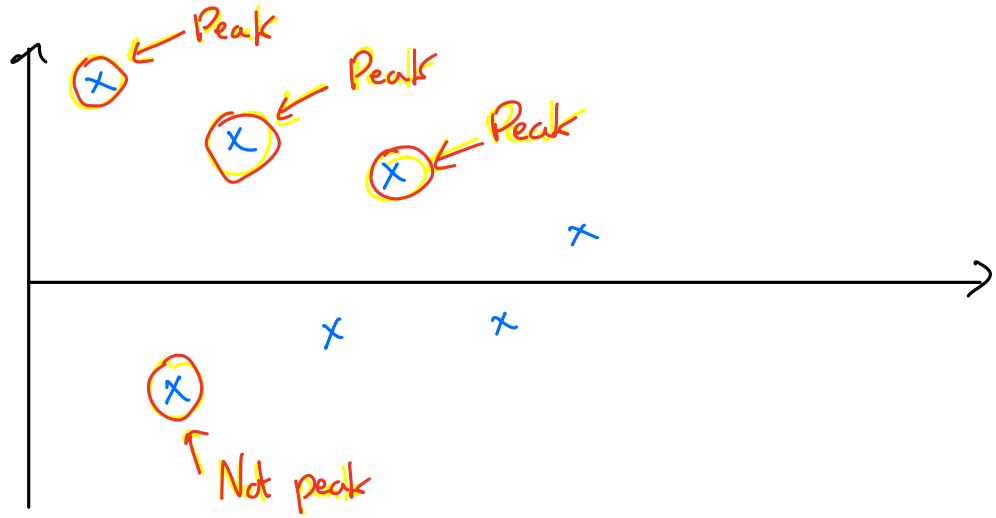


Theorem 8.6:

Every sequence contains a monotone subsequence.

Proof: We say a_n is peak if $a_n \geq a_m$ for all $m > n$.

Pic:



It must that either there are infinitely many peaks or finitely many. Assume first, there are finitely many peaks.

Then, there exists $N \in \mathbb{N}$ in which $a_N \geq a_m$ for all $m > N$. That is, a_N is the last peak.

Then letting $n_1 = N+1$, it must be that a_{n_1} is not

a peak. Hence, there exists $n_2 > n_1$, such that $a_{n_2} \geq a_{n_1}$. Similarly, there exists $n_3 > n_2 > n_1$, such that $a_{n_3} \geq a_{n_2} \geq a_{n_1}$. Doing this indefinitely defines a monotone increasing subsequence (a_{n_k}). If there are infinitely many peaks let a_{n_k} denote the k-th peak. It follows, that $a_{n_k} \geq a_{n_{k+1}}$ for all $k \in \mathbb{N}$, and thus (a_{n_k}) is monotone decreasing. ☐

Theorem 8.7 : Bolzano-Weierstrass Theorem

Every bounded sequence has a convergent subsequence.

Proof: See notes.

9 - The Cauchy Criterion

Defn 9.1: A sequence (a_n) is a **Cauchy sequence** if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$

$$|a_n - a_m| < \varepsilon$$

$n, m > N$.

Theorem 9.2: (Cauchy ⇒ Bounded)

All Cauchy sequences are bounded.

Proof: There exists an $N \in \mathbb{N}$ such that

$$|a_n - a_m| < 1$$

for all $n, m > N$. Let S be the set

$$S = \{|a_1|, |a_2|, \ldots, |a_N|, |a_{N+1}| + 1\}$$

Define $b = \max(S)$. It follows for all $m > N$

$$\begin{aligned} |a_m| &= |a_m - a_{N+1} + a_{N+1}| \\ &\leq |a_m - a_{N+1}| + |a_{N+1}| \\ &< 1 + |a_{N+1}| \\ &\leq \max(S). \end{aligned}$$

Similarly, $\max(S) \geq |a_n|$ for all $n \in \{1, \ldots, N\}$.

Hence, $|a_n| \leq \max(S) \quad \forall n \in \mathbb{N}$. ∎

Theorem 9.3: (Cauchy $\Leftrightarrow$ Convergent).

A sequence is convergent iff it is Cauchy.

Proof: ($\Rightarrow$) Suppose $\lim(a_n) = a$. Then, given $\varepsilon > 0$, there exists $N \in \mathbb{N}$ s.t

$$|a_n - a| < \varepsilon/2$$

for all $n > N$. Then, for all $n, m > N$ we have that

$$\begin{aligned} |a_n - a_m| &= |a_n - a + a - a_m| \\ &\leq |a_n - a| + |a - a_m| \\ &< \varepsilon/2 + \varepsilon/2 \\ &= \varepsilon. \end{aligned}$$

So (a_n) is Cauchy.

($\Leftarrow$) Suppose (a_n) is Cauchy. It follows that it is bounded. By Bolzano-Weierstrass, it follows there exists a convergent subsequence (a_{n_k}). Say $\lim(a_{n_k}) = a$. Then given $\varepsilon > 0$, there exists $N_1 \in \mathbb{N}$ such that

$$|a_n - a_m| < \varepsilon/2$$

for all $n, m > N_1$. Also, there exists $N_2 \in \mathbb{N}$ s.t

$$|a_{n_k} - a| < \varepsilon/2$$

for all $k > N_2$. Define $N = \max(N_1, N_2)$. Then, for all $k > N$ we have that

$$\begin{aligned} |a_k - a| &= |a_k - a_{n_{k+1}} + a_{n_{k+1}} - a| \\ &\leq |a_k - a_{n_{k+1}}| + |a_{n_{k+1}} - a| \\ &< \varepsilon/2 + \varepsilon/2 \\ &= \varepsilon. \end{aligned}$$

So (a_n) converges to a. ☐

Ex 9.4 : Let (a_n) be the sequence

$$a_{n+1} = 1 + \frac{1}{a_n},$$

and $a_1 = 1$. Then (a_n) is Cauchy and hence convergent

Since (a_n) is defined recursively, we can find the value it converges to. In this case, it converges to the Golden ratio. That is $\frac{1+\sqrt{5}}{2}$. See notes for proof.